



Approved by Chair:

Signature

COURSE SECTION INFORMATION

Big Data Tools and Techniques I Applied A.I. Solutions Development

Note: All academic inquiries will be replied to within three business days.

COURSE DESCRIPTION:

This course will provide an introduction to the current tools and frameworks used in the visualisation, streaming, and analytics of Big Data. Students will gain experience on the management of a big data project while applying current industry best practices. The focus of this course will be the creation of Big Data applications using tools such as Apache Spark and Apache Kafka.

COURSE OUTCOMES:

Upon successful completion of this course the students will have reliably demonstrated the ability to:

1. Compare and contrast the benefits and limitations of popular and widely used big data tools in big data ecosystem.
2. Build big data visualization dashboards using a variety of current industry tools.
3. Implement a big data analytics tool using Apache Spark.
4. Understand various types of databases (Understand relational and document DB)
5. Understand Hadoop
6. Design applications that implement technologies Hadoop and Apache
7. Design ETLs using various tools

LIST OF TEXTBOOKS AND OTHER TEACHING AIDS:

Reference:

Recommended Resources:

COURSE DELIVERY MODE:

Refer to the topical outline table for the delivery mode.

EVALUATION SYSTEM:

Assessment Tool:	Description:	Outcome(s) assessed:	EES assessed:	Date / Week:	% of Final Grade:
Participation	In class participation in various activities	1, 2, 3, 4, 5	1,2,3,4,5,6,7,8,9,10,11	1-3	10
Assignments	Group	1, 2, 3, 4, 5	1,2,3,4,5,6,7,8,9,10,11		50
Final Project	Group	1,2, 3, 4, 5	1,2,3,4,5,6,7,8,9,10,11		40
				TOTAL	100%

GRADING SYSTEM the passing grade for this course is: D (50%)

A +	90-100	4.0	B+	77-79	3.3	C +	67-69	2.3	D +	57-59	1.3	Below 50	F	0.0
A	86-89	4.0	B	73-76	3.0	C	63-66	2.0	D	50-56	1.0			
A-	80-85	3.7	B-	70-72	2.7	C-	60-62	1.7						

Excerpt from the College Policy on Academic Dishonesty:

The *minimal* consequence for submitting a plagiarized, purchased, contracted, or in any manner inappropriately negotiated or falsified assignment, test, essay, project, or any evaluated material will be a grade of zero on that material.

To view George Brown College policies please go to www.georgebrown.ca/policies

TOPICAL OUTLINE

Week	Topic Task	Delivery mode	Outcome(s)	Content / Activities	Resources
1	1	In Person	BIG Data Management using Apache Hadoop	- Introduction to Big Data - Overview of Big Data Ecosystem tools & frameworks - HDFS core components - Hadoop ecosystem	Resource material available on D2L
1	2	Online	Hands on practice / tutorials using Apache Hadoop	- Introduction to Relational Database - Apache Hive/HiveQL - Apache Pig	Resource material available on D2L

2	2	In Person	Hands on practice / tutorials using MongoDB Atlas	- Introduction to Document DB - MongoDB setup and introduction - Difference between Structure and semi structured - CRUD Operations using MQL	Resource material available on D2L
2	1	Online	Hands on practice / tutorials using SQL Server	- Dimensional modeling concepts - Installation SQL Serer/ MySQL - Introduction to MariaDB -Local installation / SQL Server Express	Resource material available on D2L
3	2	In Person	Hands on practice / tutorials using Microsoft Visual Studio	- SQL Basic/ Advance - ETL Concept - Installation Microsoft SSIS - Design ETL using SSIS	Resource material available on D2L
3	3	Online	Hands on practice / tutorials using Microsoft Visual Studio	- ETL Continued using SSIS	Resource material available on D2L
4	1	In Person	Hands on practice / tutorials using Jupyter Notebook	- Proof of concepts - Python with MongoDB (PyMongo) - Power Pivot	Resource material available on D2L
4	2	Online	Final Project	- Presentation for Final Project	Resource material available on D2L
Please note: this schedule may change as resources and circumstances require. For information on withdrawing from this course without academic penalty, please refer to the College Academic Calendar: http://www.georgebrown.ca/Admin/Registr/PSCal.aspx					